



Andrés Felipe Guido Montoya

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Tenth-semester student of Systems and Computer Engineering at the Universidad Nacional de Colombia. I have been part of the development team in a research project to create an educational repository management system, and I have designed and implemented interface and control systems for a seismic vibration table prototype, applying web development and multiplatform interfaces, accumulating 2 years of experience in university and research projects. Additionally, I worked 7 months at PiTransform implementing AI-based systems — working with LLMs, data pipelines, and Azure services (App Function, Web App), and building advanced RAG and information retrieval systems. I worked 6 months as a teaching assistant for the Software Engineering Project Management course. I currently work at Sento IOT, where I lead AI integration across all company processes and develop IoT data pipelines and cloud infrastructure, from data modeling and system design to production deployment.

PROFESSIONAL EXPERIENCE

Head of AI

Medellín, Colombia

Sento IOT

December 2025 – Present

- Led the company-wide AI integration strategy, embedding LLMs, RAG systems, and autonomous agents into engineering, procurement, and operational workflows.
- Designed and deployed end-to-end AI automation pipelines that significantly reduced manual effort across multiple business units, including procurement, reporting, and customer support.
- Built intelligent document and data processing systems using LLMs to automate information extraction, summarization, and structured data generation across departments.
- Implemented AI-powered workflow automations connecting internal tools, external APIs, and heterogeneous data sources into unified, intelligent pipelines.
- Conducted technical training sessions and workshops to guide cross-functional engineering teams on responsible AI adoption, prompt engineering, and best practices for production AI systems.
- Evaluated and prototyped AI models and frameworks, selecting and deploying the most suitable solutions based on business requirements, cost constraints, and scalability needs.
- Collaborated with leadership to define the company's AI roadmap, identifying high-impact automation opportunities and establishing governance standards for ethical AI adoption.
- Monitored AI system performance in production, iteratively refining models, prompts, and pipelines based on real-world data and stakeholder feedback.

Software Engineer

Medellín, Colombia

Sento IOT

May 2025 – Present

- Designed serverless microservices architecture to integrate various systems with IoT data in Ubidots.
- Designed the centralized business database.
- Implemented infrastructure on AWS.
- Implemented infrastructure as code using AWS CloudFormation.
- Implemented WhatsApp notification service for various devices.
- Implemented device query services via WhatsApp.
- Applied a customized landing page with QR code access to visualize data from transmitter devices.
- Gathered requirements and analyzed problems and solutions associated with client needs.
- Implemented CI/CD Workflows.
- Designed and implemented a serverless data pipeline on AWS (IoT Core, Lambda, S3, CloudFormation) to collect, validate, and store synchronized weight measurements and images from IoT devices, building a structured dataset for computer vision model training.

PiTransform

Full Stack Developer

Medellín, Colombia

Aug 2024 – Feb 2025

- Developed and implemented an AI-based chat system using the o3-mini model, with a particular focus on achieving high-reasoning functionalities.
- Created and deployed an advanced model capable of receiving, processing, and managing files of various formats, efficiently integrating the Intelligence Document tool for data ingestion and analysis.
- Designed and built an autonomous agent specialized in automated report generation, optimizing the processes of analysis and presentation of key information.

Teaching Assistant – Software Project Management

Universidad Nacional de Colombia

Medellín, Colombia

Aug 2024 – Feb 2025

- Contributed to the creation of educational materials.
- Supported students in their development processes with companies.
- Guided students in the implementation of Scrum and agile methodologies in their projects.
- Taught about work products focused on agile software project management.
- Reviewed and supervised the execution of student projects.

Seismic Vibration Table Prototype – Research Project

Desktop Developer

Research Assistant

Medellín, Colombia

May 2023 – April 2024

- Designed and developed a robust desktop application in Python, using asynchronous and multithreaded programming for efficient real-time data control and acquisition management.
- Implemented hardware communication and control through an Arduino controller, ensuring synchronization, stability, and integrity in the transmission of critical data.
- Led the research and standardization of scalable processes, optimizing software viability and ensuring its interoperability with specialized hardware.
- Developed specialized algorithms for the precise manipulation and processing of data, crucial for ensuring accuracy in experimental analyses.
- Thoroughly documented the project and wrote a scientific article detailing key findings and the methodology used.

EDUCATION

UNIVERSIDAD NACIONAL DE COLOMBIA

Systems and Computer Engineering.

Medellín, Colombia

October 2021 – June 2026

ADDITIONAL SKILLS

- Native Spanish speaker, English B2.
- Frontend and Full Stack Development
- Reactjs, Hooks, Merm, JWT, Axios
- Javascript, Typescript
- Nodejs, Express
- FastAPI and Python applied in various environments.
- MongoDB and SQL Databases (Intermediate).
- Application deployment on Azure, Railway, AWS and cloud services (Backblaze, Cloudflare).
- Robust authentication systems, cryptographic functions, information confidentiality management, SSH, TLS/SSL certificates, security testing (pentesting and vulnerability analysis), and Kali Linux-based tools.
- Docker
- Redis
- RAG Systems
- Azure web app
- Azure app function
- TensorFlow
- Scikit-learn
- AWS Serverless
- AWS CloudFormation
- Python
- Java
- Systems Design
- Microservices
- REST API
- Data Pipelines
- SQL
- Machine Learning
- Git / GitHub

